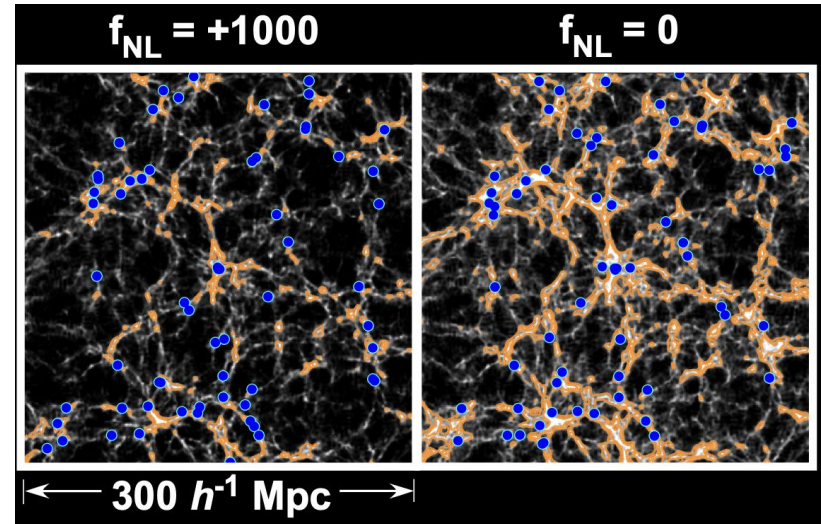
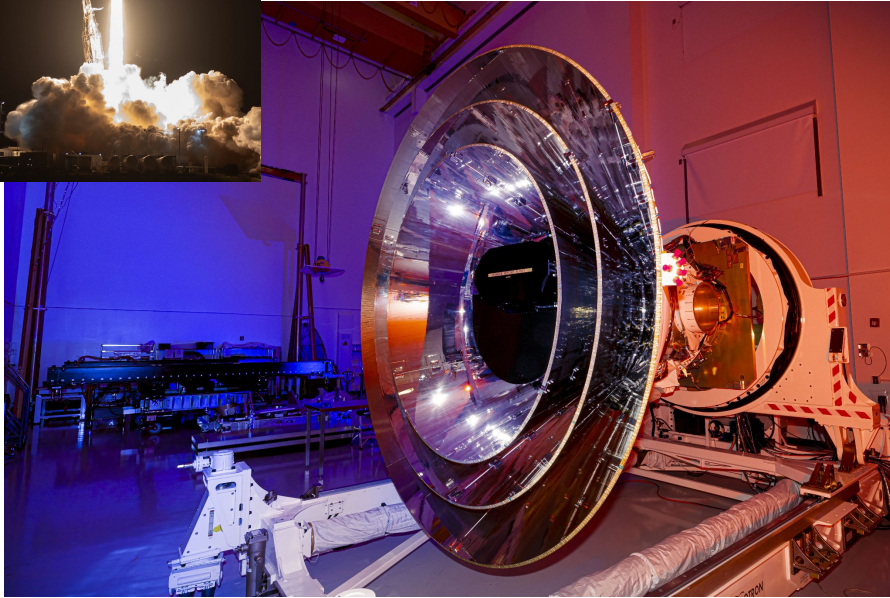




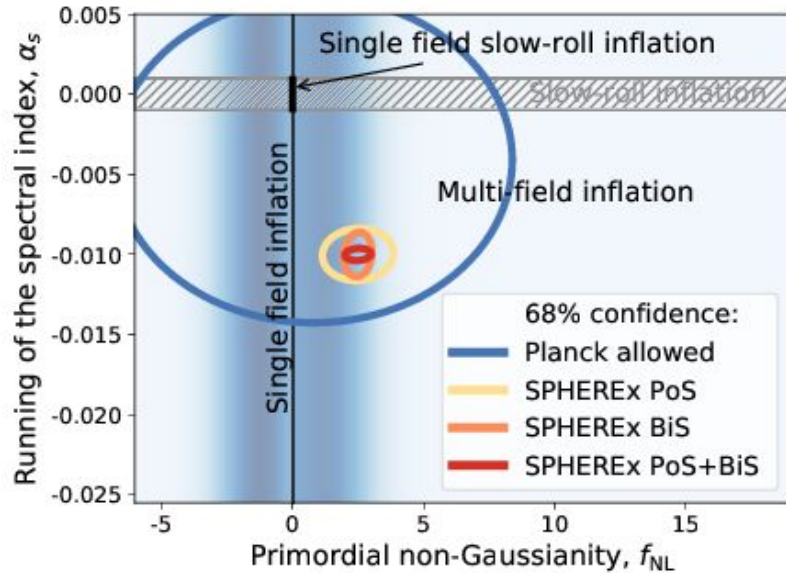
Optimizing a Neural Network Emulator for SPHEREx Galaxy Power Spectra

Grace Gibbins in collaboration with Dr. Tim Eifler, Joe Adamo, & Annie Moore

SPHEREx - Constraining Models of Inflation with f_{NL}



Primordial Non-Gaussianity: Previous Surveys



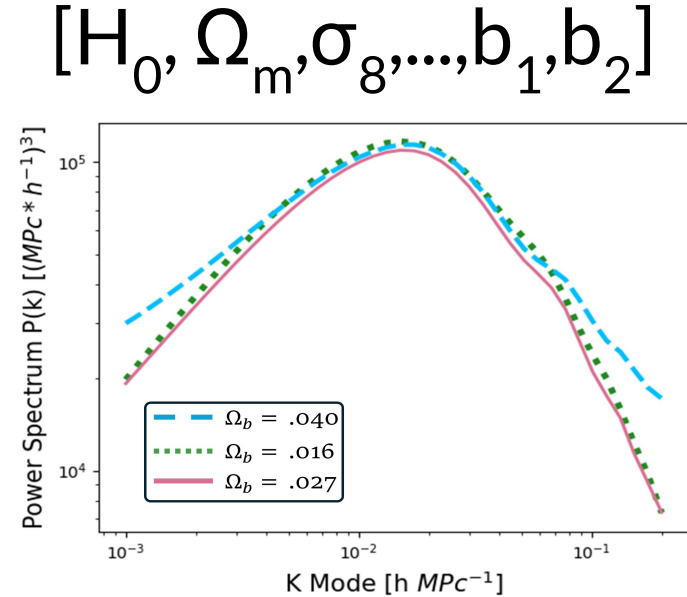
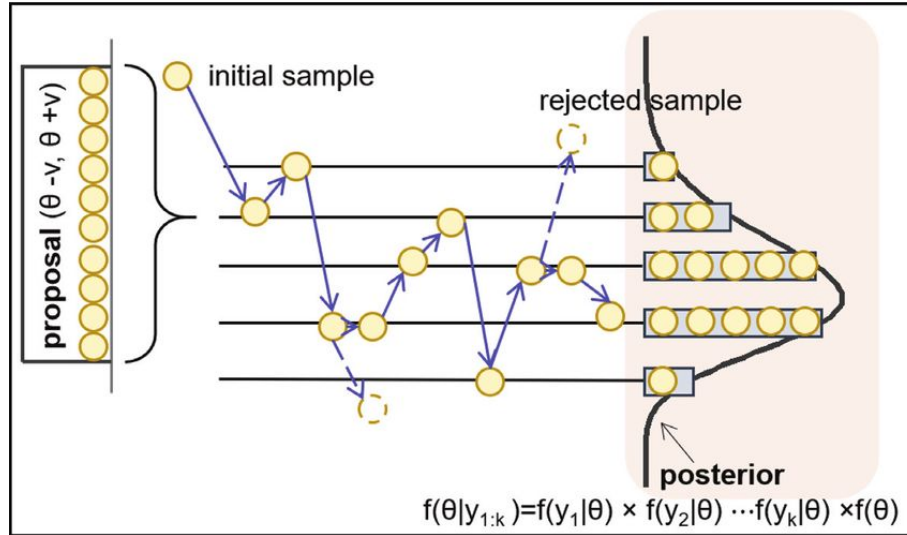
$$f_{\text{NL Planck}} \sim -0.9 \pm 20-30$$

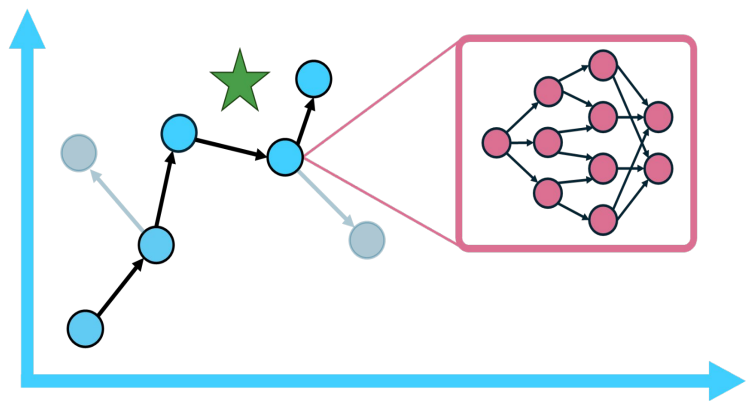
$$f_{\text{NL DESI}} \sim -3.6 \pm 7-11$$

$$\sigma_{f_{\text{NL SPHEREx}}} \sim$$

$$0.5-1$$

MCMC: Galaxy Power Spectra





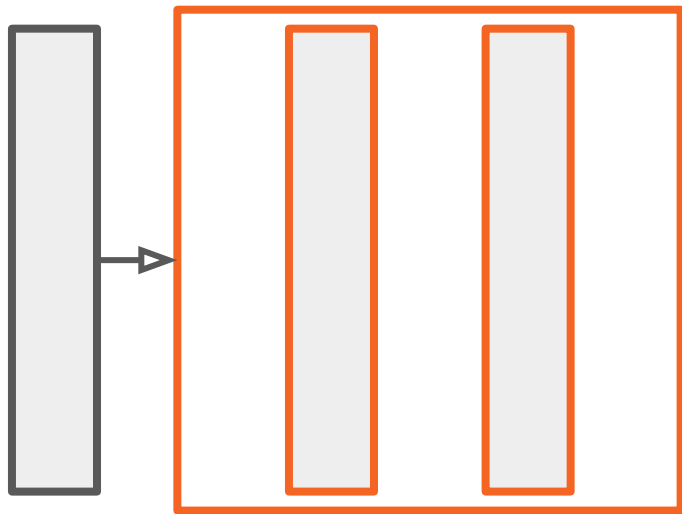
Grace Gibbins

Integrating a Neural Network into an MCMC

Accelerate model evaluation by
680x: 1s to 3ms

Input: Cosmology
& Galaxy Bias
Parameters

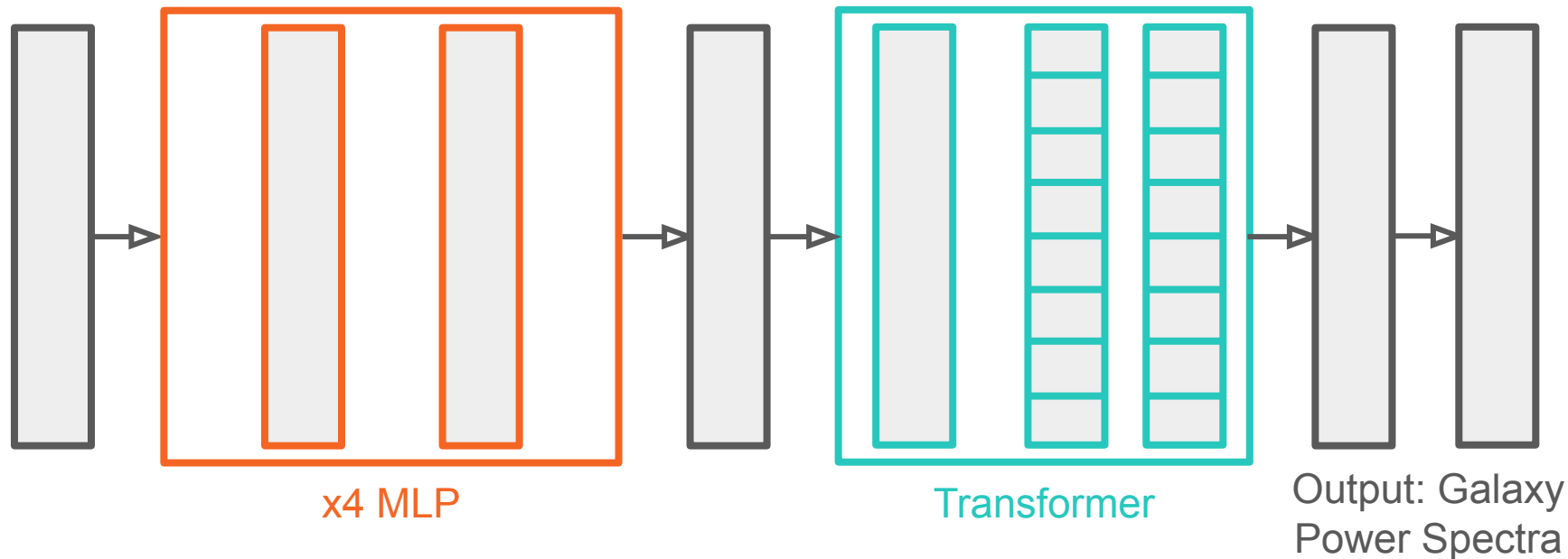
Network Architecture



x4 MLP

Input: Cosmology
& Galaxy Bias
Parameters

Network Architecture

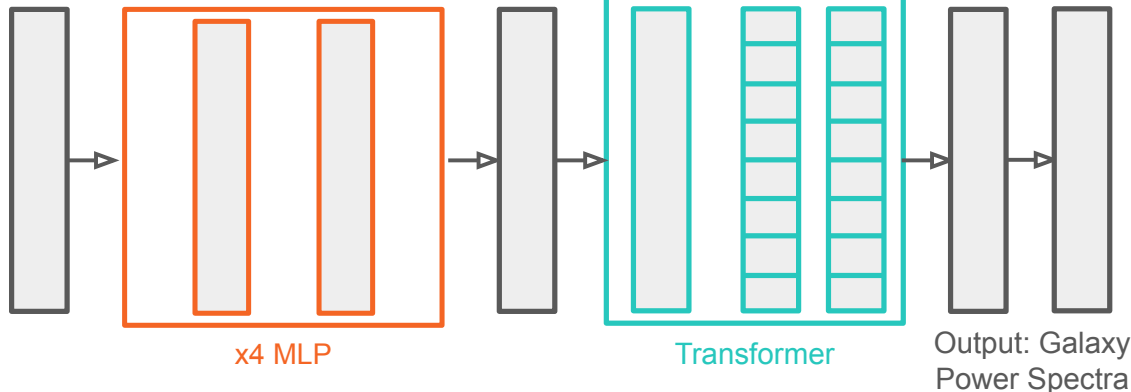


Optimizing the Emulator: Network Hyperparameters

- The amount at which the neural network allows the model to change its weights at each step.

Hyperparameters
Learning Rate
Batch Size
MLP Blocks
Layers / Block
Transformers

Input: Cosmology &
Galaxy Bias Parameters

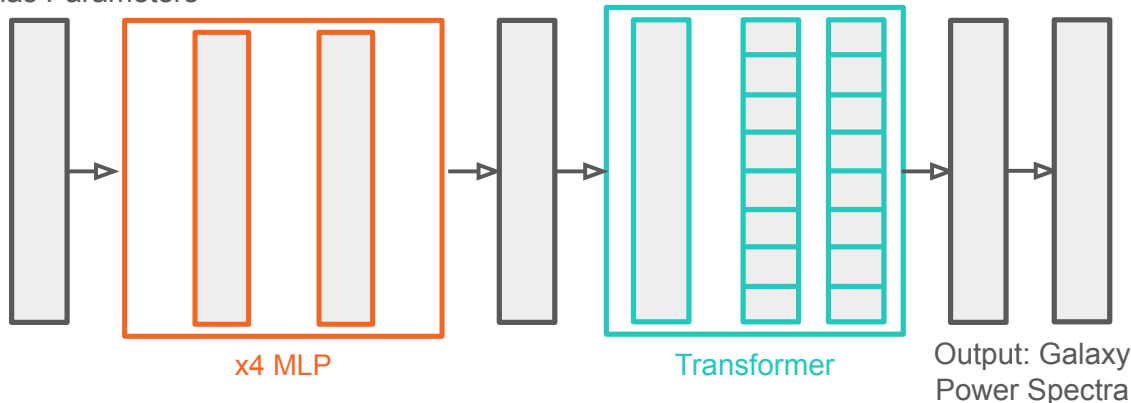


Optimizing the Emulator: Network Hyperparameters

- The number of training samples the network trains on at one time.

Hyperparameters
Learning Rate
Batch Size
MLP Blocks
Layers / Block
Transformers

Input: Cosmology &
Galaxy Bias Parameters

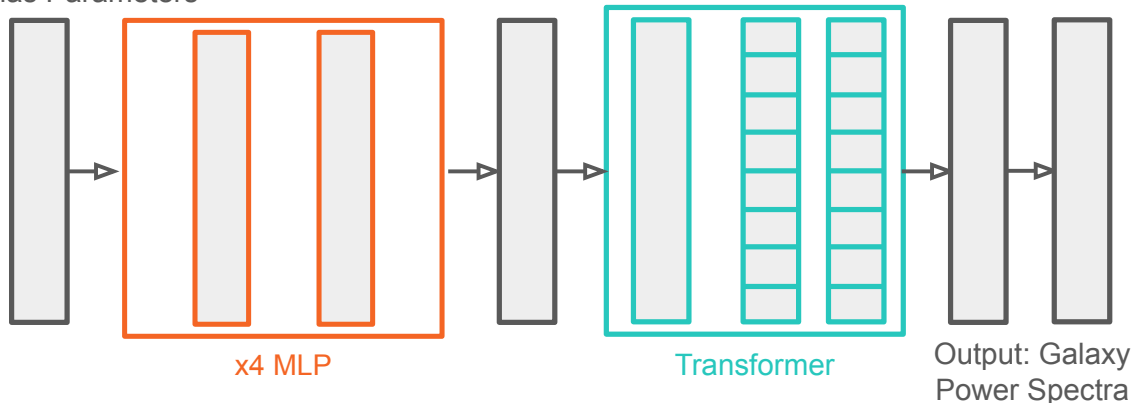


Optimizing the Emulator: Network Hyperparameters

Hyperparameters
Learning Rate
Batch Size
MLP Blocks
Layers / Block
Transformers

- Number of blocks of layers that input the data, create a weighted sum, and add nonlinearity.

Input: Cosmology &
Galaxy Bias Parameters

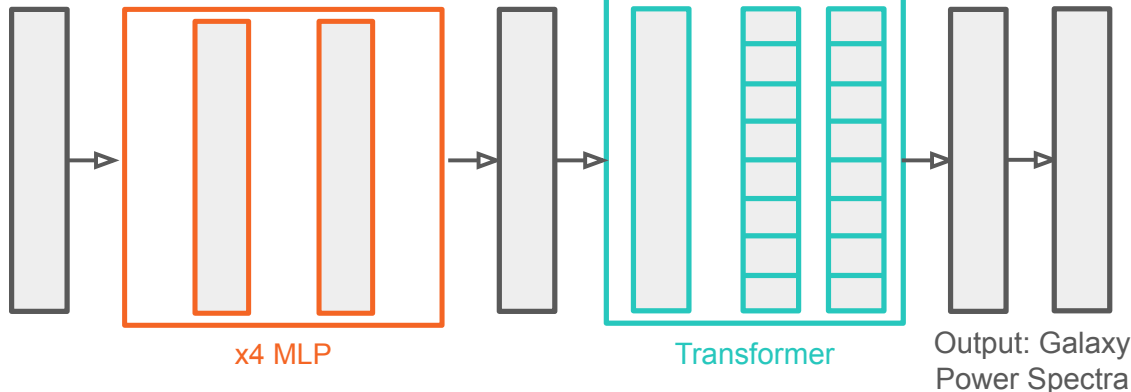


Optimizing the Emulator: Network Hyperparameters

- Number of hidden layers of neurons in each MLP block.

Hyperparameters
Learning Rate
Batch Size
MLP Blocks
Layers / Block
Transformers

Input: Cosmology &
Galaxy Bias Parameters

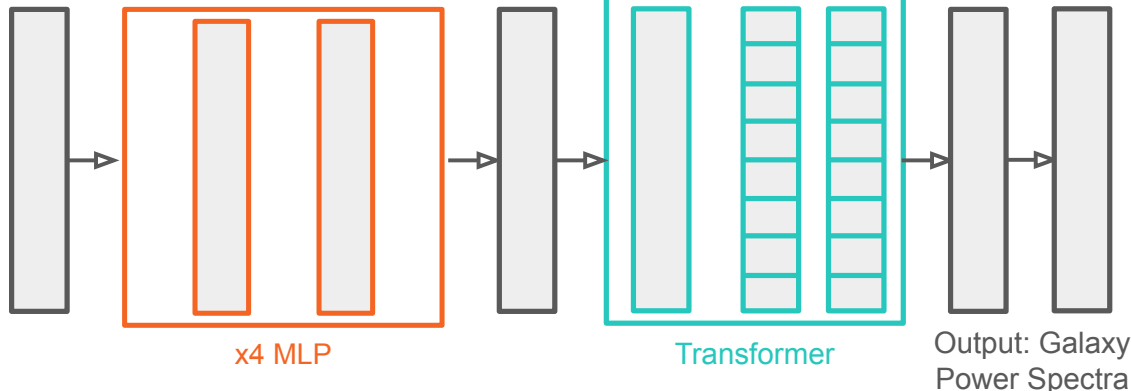


Optimizing the Emulator: Network Hyperparameters

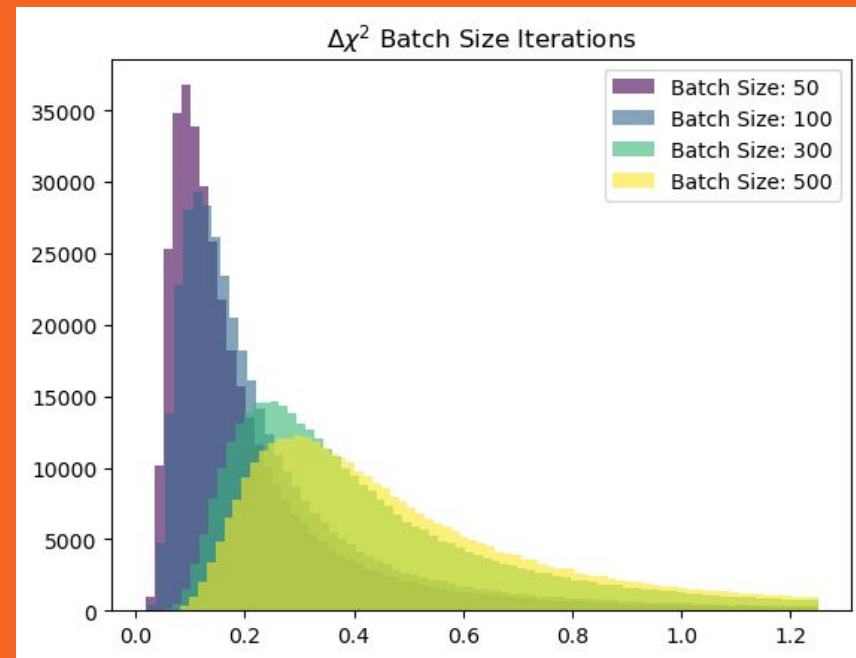
- Number of blocks similar to those used by LLM that try to interpret the relationship between input elements.

Hyperparameters
Learning Rate
Batch Size
MLP Blocks
Layers / Block
Transformers

Input: Cosmology &
Galaxy Bias Parameters



Optimizing the Network: Picking Hyperparameters





Summary

- SPHEREx: All sky spectrophotometric survey.
- Science Goal: Measure primordial non gaussianity, f_{NL} , from galaxy power spectra.
- Project Goal: Accelerate forecasts by integrating a neural network emulator into the MCMC for parameter impact studies.
- Results: 680x speed up compared to traditional EFT code (1 second to 3 milliseconds).

Extra Slide: Hyperparameter Values

Hyperparameter	Default Value	Iteration Values	Final Value
Learning Rate	0.001	0.0001, 0.0005, 0.0025, 0.005, 0.0075, 0.01	0.005
Batch Size	200	50, 100, 300, 500	200 ^a
MLP Blocks	2	1, 3, 4, 5	4
Layers / Block	4	2, 3, 5, 6	2
Transformer Blocks	1	2, 3, 4, 5	1 ^a
